

PrimeTrade.ai Hiring Assignment

Bitcoin Market Sentiment vs Hyperliquid Trader Performance Analysis

Executive Summary

This report analyzes the relationship between Bitcoin market sentiment and trader performance using the Bitcoin Fear & Greed Index and Hyperliquid historical trading data. After cleaning and merging the datasets, 211,224 trading records were analyzed. The objective was to determine how market sentiment influences profitability, win rates, trading activity, and trader behavior.

Dataset Overview

Dataset 1: Bitcoin Fear & Greed Index containing sentiment classifications (Extreme Fear, Fear, Neutral, Greed, Extreme Greed). Dataset 2: Hyperliquid historical trading data containing account activity, execution price, trade size, side, fees, and closed profit/loss metrics. The datasets were merged using trading date.

Data Preparation & Merge Process

Date fields were standardized and converted to datetime format. Trading records were mapped to the corresponding daily sentiment classification. The merged dataset contained 211,224 records, with only six unmatched rows (<0.01%), which were removed. Additional features such as win/loss indicators were created for analysis.

Exploratory Analysis

The analysis focused on profitability by sentiment, win rates, trading volume, buy/sell behavior, transaction fees, and top-performing traders. Aggregations and visualizations were used to compare trader performance across different market sentiment regimes.

Key Findings

1. Extreme Greed delivered the highest average Closed PnL (67.89) and the highest win rate (46.49%). 2. Fear periods generated the highest aggregate profit (~3.36M) and the highest trading volume (~483M USD). 3. Extreme Fear produced the lowest win rate (37.06%), indicating challenging trading conditions. 4. Sell orders slightly exceeded buy orders across most sentiment categories. 5. Fear periods incurred the highest average fees, suggesting elevated trading activity and market participation.

Trading Recommendations

- Increase exposure during strong bullish sentiment while maintaining risk controls.
- Reduce leverage and tighten risk management during Extreme Fear conditions.
- Monitor Fear regimes closely because they generated the highest total profit opportunities.
- Incorporate sentiment indicators into position sizing and portfolio allocation decisions.
- Optimize trade frequency during high-activity periods to reduce fee impact.

Conclusion

The results indicate a meaningful relationship between market sentiment and trading outcomes. Trader profitability, win rates, volume, and behavior all varied across sentiment regimes. These

findings support the use of sentiment-aware trading strategies for improved decision-making and risk management.

charts

